Lecture 1: Introduction and Working with Data

Vegard Høghaug Larsen

GRA 4160: Predictive Modelling with Machine Learning

January 7th 2026

About me

Vegard Høghaug Larsen
(Department of Data Science and Analytics)

Background:
PhD in Economics followed by six years in Norges Bank.

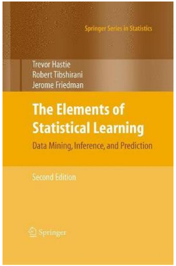
Contact:
Email: vegard.h.larsen@bi.no
Office: B3Y-075

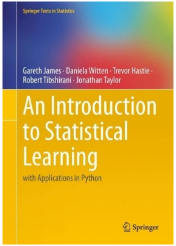
Research
My work is at the intersection of economics and data science, where I use machine learning and natural language processing techniques to study the transmission of economic shocks, understand how agents form their expectations, and develop methods to measure unobserved concepts such as sentiment, uncertainty, and climate risk.

Syllabus

Primary Textbook:
The Elements of Statistical Learning
by Hastie, Tibshirani, & Friedman
([Free PDF Link])

Gentle Introduction:
An Introduction to Statistical Learning
by James, Witten, Hastie, & Tibshirani
([Free PDF Link])





Prerequisites / What I Expect You to Know

Foundations

- **Mathematics:** Basic linear algebra and multivariate calculus.
- **Statistics:** Fundamental probability and statistical concepts.

Prerequisites / What I Expect You to Know

Foundations

- **Mathematics:** Basic linear algebra and multivariate calculus.
- **Statistics:** Fundamental probability and statistical concepts.

Technical Skills

- **Python Programming:** Experience with NumPy, Pandas, and Matplotlib.
- **OOP:** Ability to write and understand Object-Oriented structures.

About this Course

- Focused on **predictive modeling** using traditional (pre-deep learning) methods.

About this Course

- Focused on **predictive modeling** using traditional (pre-deep learning) methods.
- **Core Algorithms:** Linear/logistic regression, decision trees, and ensemble methods.

About this Course

- Focused on **predictive modeling** using traditional (pre-deep learning) methods.
- **Core Algorithms:** Linear/logistic regression, decision trees, and ensemble methods.
- **Hands-on Practice:** Writing code from scratch to understand mechanics.

About this Course

- Focused on **predictive modeling** using traditional (pre-deep learning) methods.
- **Core Algorithms:** Linear/logistic regression, decision trees, and ensemble methods.
- **Hands-on Practice:** Writing code from scratch to understand mechanics.
- **Real Workflows:** Preprocessing, training, tuning, and evaluation.

About this Course

- Focused on **predictive modeling** using traditional (pre-deep learning) methods.
- **Core Algorithms:** Linear/logistic regression, decision trees, and ensemble methods.
- **Hands-on Practice:** Writing code from scratch to understand mechanics.
- **Real Workflows:** Preprocessing, training, tuning, and evaluation.
- **Goal:** Critical evaluation and improvement of models for real-world tasks.

Mini Project

Logistics

- **Group Work:** 2–4 students (Form groups by next week).
- **Objective:** Apply methods to a real-world problem.
- **Feedback:** Up to 2 scheduled sessions per group.
- **GitHub:** Recommended for sharing code/report.

Evaluation

- **Weight:** 30% of final grade.
- **Presentation:** End of semester (External grader).
- **Exam:** Project topics are relevant for the final exam.

Course Structure

- **Interactive Format:** Minimal lecture-style; focus on live coding.

Course Structure

- **Interactive Format:** Minimal lecture-style; focus on live coding.
- **Typical Session:**
  1. Brief concept intro (slides)
  2. Live coding & examples
  3. Exercises

Course Structure

- **Interactive Format:** Minimal lecture-style; focus on live coding.
- **Typical Session:**
  1. Brief concept intro (slides)
  2. Live coding & examples
  3. Exercises
- **Materials:** Background notebooks + Exercise notebooks.

Course Structure

- **Interactive Format:** Minimal lecture-style; focus on live coding.
- **Typical Session:**
  1. Brief concept intro (slides)
  2. Live coding & examples
  3. Exercises
- **Materials:** Background notebooks + Exercise notebooks.
- **Expectation:** Review notebooks *before* class.

Using Generative AI as a Learning Tool

- **Role:** Use GenAI as a learning partner, not a replacement for reasoning.

Using Generative AI as a Learning Tool

- **Role:** Use GenAI as a learning partner, not a replacement for reasoning.
- **Usage:** Ask for help with implementing code, clarifications, alternative solutions, and code explanations.

Using Generative AI as a Learning Tool

- **Role:** Use GenAI as a learning partner, not a replacement for reasoning.
- **Usage:** Ask for help with implementing code, clarifications, alternative solutions, and code explanations.
- **Tools:**
  - **Google Gemini:** Access to BI-students.
  - **GitHub Copilot:** Free for students, integrated with VSCode.

Using Generative AI as a Learning Tool

- **Role:** Use GenAI as a learning partner, not a replacement for reasoning.
- **Usage:** Ask for help with implementing code, clarifications, alternative solutions, and code explanations.
- **Tools:**
  - **Google Gemini:** Access to BI-students.
  - **GitHub Copilot:** Free for students, integrated with VSCode.
- **Warning:** Always apply critical thinking. You are responsible for the output.

Tools: Integrated Development Environment (IDE)

Why use an IDE?

- Integrated tools (linting, debugging).
- Higher productivity.

Recommendation: VS Code

- Lightweight & Cross-platform.
- Excellent Python & Jupyter support.
- Built-in Git version control.

VS Code is industry standard and free.

Let's get to know you.

- Name and academic/professional background?
- Career goals in Data Science/ML?
- Main learning objectives for this course?
- Questions/Concerns?

Plan for Today

1. Working with Data in Jupyter
   - **Why Jupyter?** The standard for prototyping.
   - **Preprocessing:** Why cleaning, encoding, and scaling are mandatory.
   - **Reproducibility:** Systematic workflows in Python.
2. Hands-on Material
   - Lecture Notebook: `01_Working_with_data_in_jupyter...ipynb`
   - Exercise Notebook: `01_Data_preprocessing_titanic.ipynb`

The Machine Learning Workflow

1. **Define & Collect:** Clarify task, gather data.

The Machine Learning Workflow

1. **Define & Collect:** Clarify task, gather data.
2. **Preprocessing:** Clean, transform, encode.

The Machine Learning Workflow

1. **Define & Collect:** Clarify task, gather data.
2. **Preprocessing:** Clean, transform, encode.
3. **Model Selection:** Choose algorithm (Linear, Tree, etc.).

The Machine Learning Workflow

1. **Define & Collect:** Clarify task, gather data.
2. **Preprocessing:** Clean, transform, encode.
3. **Model Selection:** Choose algorithm (Linear, Tree, etc.).
4. **Training & Tuning:** Fit model, optimize hyperparameters.

The Machine Learning Workflow

1. **Define & Collect:** Clarify task, gather data.
2. **Preprocessing:** Clean, transform, encode.
3. **Model Selection:** Choose algorithm (Linear, Tree, etc.).
4. **Training & Tuning:** Fit model, optimize hyperparameters.
5. **Evaluation:** Test on unseen data, deploy.

Traditional ML vs. Deep Learning

Aspect	Traditional ML	Deep Learning
Features	Manual engineering (domain knowledge)	Automated extraction (representation learning)
Data Needs	Small/Medium structured data	Large datasets; Unstructured (Images/Text)
Complexity	Lower (Linear, Trees, SVM)	High (Deep Neural Networks)
Interpretability	Transparent (White-box)	Opaque (Black-box)

Overfitting vs. Underfitting

Underfitting (High Bias)

- Model is too simple.
- Misses underlying trends.
- **Fix:** More complex model, new features.

Overfitting (High Variance)

- Fits noise, not signal.
- Fails on unseen data.
- **Fix:** Regularization, more data, simpler model.

Train, Validation, and Test Splits

- **Training Set (60-80%):** Fit model parameters.
- **Validation Set (10-20%):** Hyperparameter tuning & model selection.
- **Test Set (10-20%):** Final evaluation on "unseen" data.

Crucial: Use random splits or stratification to ensure representative samples.

Why Genuine Learning Matters

You have unlimited resources (AI, Home Exam). Why learn the math?

- **Skill Building:** Solve unseen problems tomorrow, not just today's.
- **Critical Thinking:** Detect AI hallucinations.
- **Career:** Employers need people who can *fix* AI, not just use it.
- **Exam Rigor:** You must **justify** every step. "The AI said so" is not a valid answer.

Final Expectations

- Engage actively.
- Be ready to talk through your solutions.
- **Your genuine knowledge is what matters.**